

Solomon Yamoah Effah

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[Google Scholar](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

Ph.D. Candidate in Physical Chemistry, Wayne State University	2021–present
Dissertation: Computational Strategies for Drug Discovery and Fluorescent Molecular Probe Design	
Advisor: Prof. Alice R. Walker	
M.A. in Chemistry, Wayne State University	August 2024
B.S. in Chemistry, First Class Honors, Kwame Nkrumah University of Science and Technology, Ghana	August 2020

RESEARCH EXPERIENCE

Graduate Research Scientist, Walker Lab, Wayne State University	August 2024–present
Research programs: computational design of perylene-modified fluorescent nucleic acid probes for molecular imaging; structure-based discovery of tobramycin-derived heparanase inhibitors; excited-state QM/MM dynamics of photoactive systems.	
Graduate Research Assistant, Walker Lab, Wayne State University	August 2023–April 2024
Graduate Teaching Assistant, Department of Chemistry, Wayne State University	September 2021–May 2024
General Chemistry and Physical Chemistry laboratory courses.	
Teaching Assistant, Department of Chemistry, KNUST, Kumasi, Ghana	September 2020–July 2021
Undergraduate Researcher, Theoretical and Computational Chemistry Laboratory, KNUST	May 2019–August 2020

PEER-REVIEWED PUBLICATIONS

4. Jiusheng Lin, Oksana Gerlits, Daniel W. Kneller, Kevin L. Weiss, Leighton Coates, Jennifer L. Clarke, Mark A. Hix, Solomon Y. Effah, et al. Environmental contributions to proton sharing in protein low-barrier hydrogen bonds. *Biochemistry*, 2025. Featured on the front cover. DOI: [10.1021/acs.biochem.5c00762](https://doi.org/10.1021/acs.biochem.5c00762)
3. Michael Hotor, Joseph Wakpal, Solomon Y. Effah, Nur-E Alom, Alice R. Walker, and Hien M. Nguyen. Could hydrophobicity of sulfated pseudo-trisaccharides derived from repurposing aminoglycoside tobramycin modulate the enzymatic activity of heparanase? *Journal of Medicinal Chemistry*, 68(12):12708–12732, 2025. DOI: [10.1021/acs.jmedchem.5c00611](https://doi.org/10.1021/acs.jmedchem.5c00611)
2. Solomon Y. Effah, Mark A. Hix, and Alice R. Walker. Strategic design of fluorescent perylene-modified nucleic acid monomers: Position-, phosphorylation-, and linker-dependent control of electron transfer. *Journal of Chemical Information and Modeling*, 65(6):2940–2949, 2025. DOI: [10.1021/acs.jcim.4c02223](https://doi.org/10.1021/acs.jcim.4c02223)
1. Solomon Yamoah Effah, W. K. D. N. Kaushalya, Mark A. Hix, and Alice R. Walker. Computational investigations of the excited state dynamics and quenching mechanisms of polycyclic aromatic hydrocarbon DNA adducts in solution. *Electronic Structure*, 4(4):044003, 2022. DOI: [10.1088/2516-1075/ac8c16](https://doi.org/10.1088/2516-1075/ac8c16)

CONFERENCE PROCEEDINGS AND PUBLISHED ABSTRACTS

1. Jiusheng Lin, Oksana Gerlits, Daniel W. Kneller, Kevin L. Weiss, Leighton Coates, Mark A. Hix, Solomon Y. Effah, et al. Interrogating low barrier hydrogen bonds with neutrons, X-rays, and computation. *Structural Dynamics*, 12(5_Supplement):A323, 2025. DOI: [10.1063/4.0000308](https://doi.org/10.1063/4.0000308)

MANUSCRIPTS IN PREPARATION

- Solomon Y. Effah, Michael Hotor, Hien M. Nguyen, and Alice R. Walker. Beyond hydrophobic landscapes: A computational atlas of tobramycin-based heparanase inhibitors with selectivity over platelet factor 4 chemokine.
- Solomon Y. Effah, J. Bour, and Alice R. Walker. Development of red-shifted perylene fluorophores via push-pull substitution: A computational and synthetic approach.

CONFERENCE PRESENTATIONS

- "Hydrophobic modifications of tobramycin-based sulfated oligosaccharides enhance heparanase inhibition and selectivity." Oral. ACS Spring 2026, Division of Medicinal Chemistry, Atlanta, GA. March 2026.
- "Ultrafast charge transfer dynamics in perylene-modified nucleic acids: Computational design principles for enhanced DNA imaging." Poster. ACS Spring 2026, Division of Physical Chemistry, Atlanta, GA. March 2026.
- "Controlling fluorescence in DNA probes: Structural design of perylene-modified nucleic acids for enhanced imaging." Poster. 52nd NOBCCHE Annual National Conference. 2025.
- "Computational rational design of nucleic acid and carbohydrate drug scaffolds." Invited seminar. Analytical–Physical Seminar Series, Wayne State University. March 2025.
- "Computational investigation of the photophysical properties of perylene bound to nucleobase analogues." Poster. ACS Fall 2023, Division of Computers in Chemistry.
- Poster. 50th NOBCCHE Annual National Conference, New Orleans, LA. September 2023.

HONORS AND AWARDS

Winner, Schrödinger "Molecules & Models" Virtual Graduate Science Fair (2026)

Dan Trivich Memorial Award for Research in Physical Chemistry, Wayne State University (2026)
Sigma Xi, The Scientific Research Honor Society — Full Member, elected by nomination (2026)
Rumble Fellowship / A. Paul and Carole C. Schaap Endowed Distinguished Graduate Award, Wayne State University (2025–2026)
3M RISE Alumnus (2025)
First Place Poster Presentation, Wayne State University Graduate School Symposium
Departmental Citation for Excellence in Teaching Service, Department of Chemistry, Wayne State University
Best Graduating Student, Department of Chemistry, KNUST, Ghana (2020)
NOBCChE Advancing Science Conference Grant, Tier 1 (2023, 2025)

PEER REVIEW AND EDITORIAL SERVICE

Associate Editor, International Journal of Photochemistry and Photobiology
IOP Trusted Reviewer, Institute of Physics Publishing
Journal referee: Journal of Physics: Condensed Matter, ACS Omega
Abstract reviewer, 52nd NOBCChE Annual National Conference Technical Sessions (2025)
ACS Reviewer Lab — Certified (2025)

TECHNICAL EXPERTISE

Methods: Molecular dynamics, DFT/TD-DFT, QM/MM Born–Oppenheimer excited-state dynamics, MM-PBSA/MM-GBSA binding free energies, docking and virtual screening, pharmacophore modeling, QSAR/QSPR, ADMET prediction.
Software and computing: AMBER, Schrödinger Suite, Gaussian, NAMD, VMD, AutoDock, MOE, PyMOL; Python (NumPy, Pandas, SciPy, RDKit, scikit-learn), Bash, Git; Linux/SLURM and GPU-accelerated computing (CUDA).

PROFESSIONAL MEMBERSHIPS AND LEADERSHIP

American Chemical Society — Member, Divisions of Medicinal Chemistry, Physical Chemistry, and Computers in Chemistry
NOBCChE — Vice President, Wayne State University Chapter (2024–2025); Event Coordinator (2023–2024)
Afrisnet (African Network for Graduate Students and Postdoctoral Researchers) — General Secretary (2022–2023)
Ghana Students' Chemical Society, KNUST Chapter — Academic Board Chairman (2018–2019)

TRAINING AND PROFESSIONAL DEVELOPMENT

FutureGRIN NextGen Computer-Aided Drug Design Training Program
62nd Hands-On Workshop on Computational Biophysics, Auburn University (2024)
MolSSI Machine Learning and Chemistry Workshop; American Conference in Theoretical Chemistry (2024)

SERVICE AND OUTREACH

STEM mentor and volunteer, Motor City STEAM Foundation, Detroit (2025)
NOBCChE Wayne State University Chapter STEM mentoring programs (2023–2025)
Chemistry demonstrations for K–5 students, Chrysler Elementary School, Detroit (2023)